



## **Subject: Database Management System (BCAC0020)**

### **Assignment –2 Program BCA**

**Submission Date: 30/04/24**

**Q1.** A task of analyzing database is given to the Database Administrator of the company having the following relational database and give an expression in relational algebra each of the following queries:

Employee (person-name, street, city)

Works (person-name, company-name, salary)

Company (company-name, city)

Managers (person-name, manager-name)

- Find the names of all employees who work for First Bank Corporation.
- Find the names and cities of residences of all employees who work for First Bank Corporation.
- Find the names of all the employees who do not work for First Bank Corporation.
- Find names of all employees who earn more than \$10000 per annum.
- Find names of all employees who earn more than every employee of Small Bank Corporation.

**Q2.** Consider the following schema:

Suppliers (sid : integer, sname : string, address : string)

Parts (pid : integer, pname : string, color : string)

Catalog (sid : integer, pid : integer, cost : real)

The key fields are underlined and domain of each field is listed after the field name. Give an expression in SQL for each of the following queries:

- Find the name of suppliers who supply some red parts
- Find the sids of suppliers who supply some red or green parts
- Find the sids of suppliers who supply some red part or are at 221 packer Av

**Q3.** Consider a relation  $R(A,B,C,D,E,F)$  and set of all FDs on  $R$  as given below:  $\{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, F \rightarrow B\}$ . Convert this relation in 2NF.

**Q4.** Consider a relation  $R(A,B,C,D,E,F)$  and set of all FDs on  $R$  as given below:  $\{AB \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow F, F \rightarrow A\}$ . Check whether this relation is in 3NF or not. If not then convert it into 3NF.

**Q5.** What is transaction? Explain ACID properties of transactions.

**Q6.** Consider the following two schedules and check which one is conflict serializable and which one is not:

**S1:** R1(x), R1(y), R2(x), R2(y), W2(y), W1(x)

**S2:** R1(x), R2(x), R2(y), W2(y), R1(y), W1(x)

**Q7.** Which of the following schedules follow the principles of Two Phase Locking? Explain with reason.

**S1:** L1 (X), L1 (Y), L2 (Z), U2 (Z), U1 (Y), U1 (X)

**S2:** L1 (X), U1 (X), L1 (Y), L2 (Z), U2 (Z), U1 (Y)

**S3:** L1 (X), U1 (X), L1 (Y), U1 (Y), L2 (Z), U2 (Z)

**S4:** L1 (X), L1 (Y), L2 (Z), U2 (Z), U1 (Y), U1 (X), L2 (X), U2 (X)

Use the following notation for this question:

$L_i(X)$  denotes transaction  $i$  acquires lock on data item  $X$ , and

$U_i(X)$  denotes transaction  $i$  releases lock on data item  $X$

**Q8:** Explain the role of Data Base Administrator in terms of DBMS.